

Quadratics & Nonlinear Equations



Solving quadratics by factoring

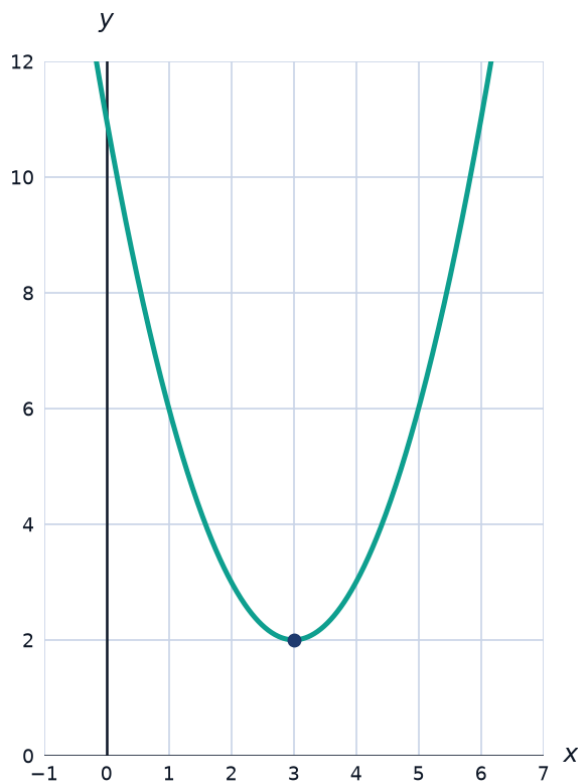
- 1 Solve $x^2 - 5x + 6 = 0$ by factoring.
 - 2 Solve $2x^2 + 7x - 4 = 0$ by factoring.
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Solving quadratics with the quadratic formula

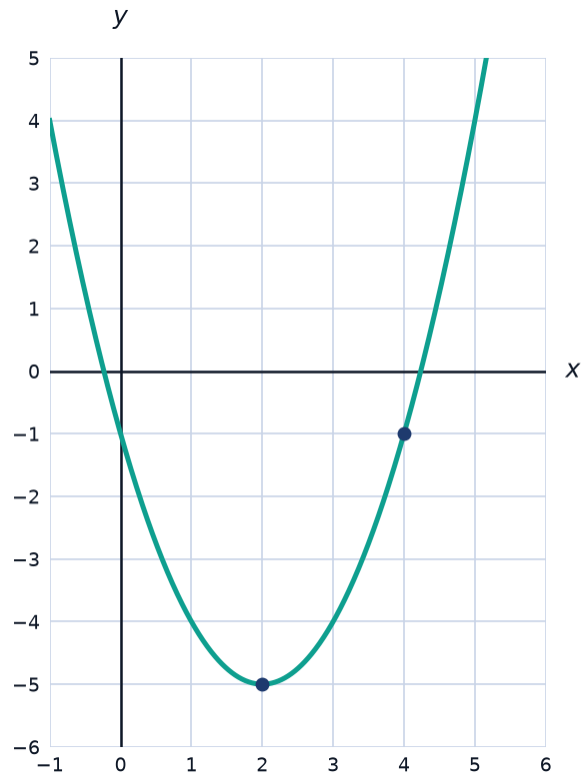
- 3 Solve $x^2 + 4x - 1 = 0$ using the quadratic formula, giving exact answers.
 - 4 Use the discriminant to determine the number of real solutions of $3x^2 - 2x + 5 = 0$, without solving the equation.
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Completing the square and vertex form

- 5 Write $f(x) = x^2 - 6x + 11$ in vertex form by completing the square, and state the vertex.

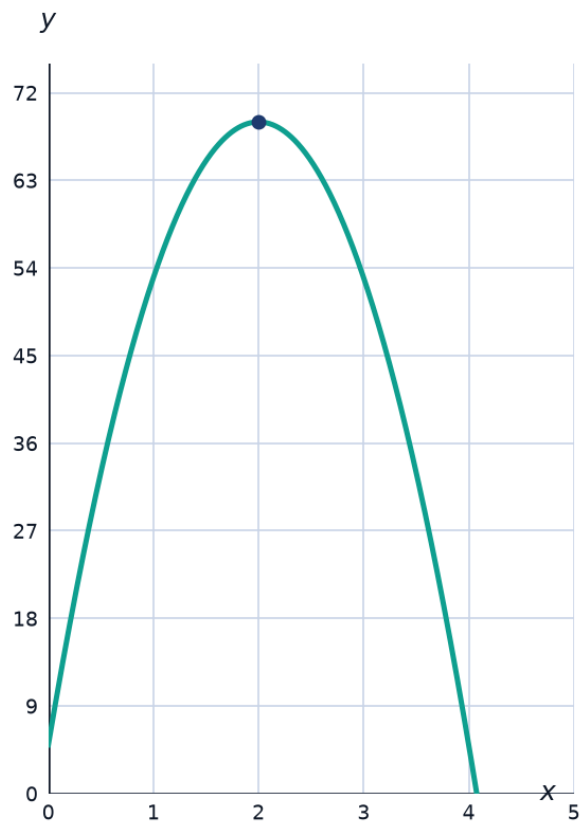


- 6 A parabola has vertex $(2, -5)$ and passes through $(4, -1)$. Find its equation in vertex form.

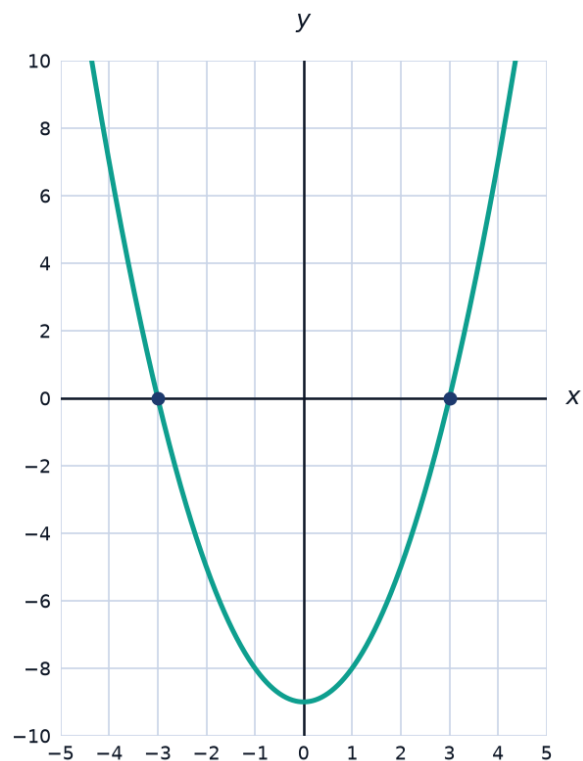


Finding zeros, maximum, and minimum values

- 7 A ball's height is modeled by $h(t) = -16t^2 + 64t + 5$, where t is time in seconds. Find the maximum height and the time at which it occurs.

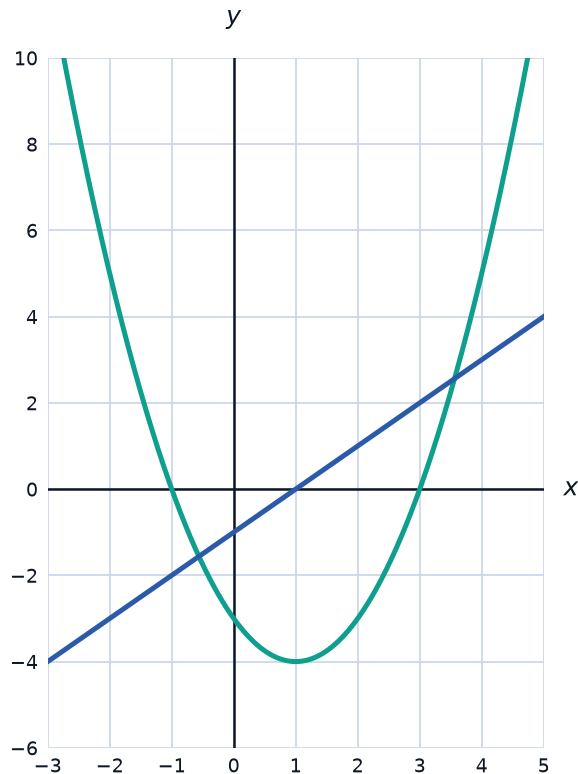


- 8 Find the zeros of $f(x) = x^2 - 9$, and interpret them as x -intercepts on a graph.



Systems with one linear and one quadratic equation

- 9 Solve the system: $y = x^2 - 2x - 3$ and $y = x - 1$.



Exponential growth and decay

- 10 A population of bacteria triples every hour, starting with 200 bacteria. Write an exponential model and find the population after 4 hours.
- 11 A car worth 24000 dollars depreciates by 15 percent each year. Write an exponential model and find its value after 3 years, to the nearest dollar.

Radical and rational equations

- 12 Solve $\sqrt{2x + 3} = 5$, and check your solution.
- 13 Solve $\frac{3}{x - 2} = \frac{1}{x + 1}$, and state any restrictions on x .
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Nonlinear relationships in context

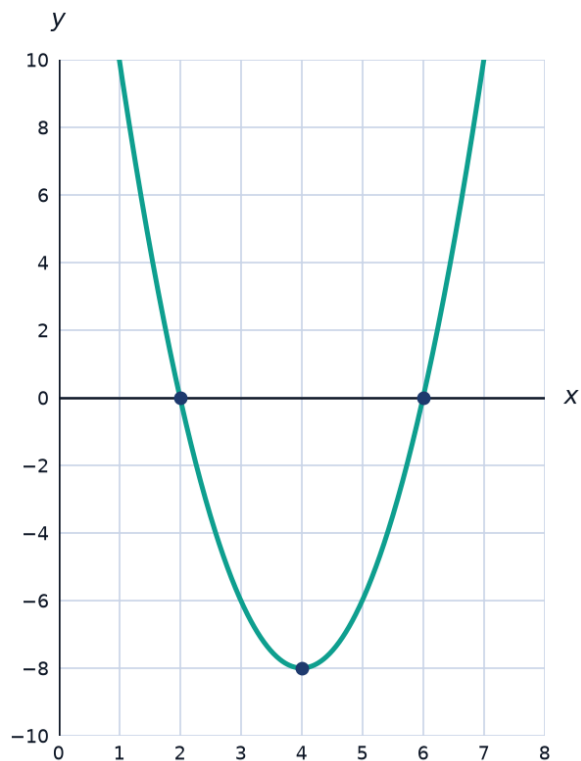
- 14 The area of a square garden is modeled by $A(s) = s^2$, where s is the side length in meters. If the area must be at most 144 square meters, find the maximum possible side length.

Factoring special products

- 15 Factor $9x^2 - 25$ completely.
- 16 Factor $x^2 + 10x + 25$ completely, and identify the special pattern used.

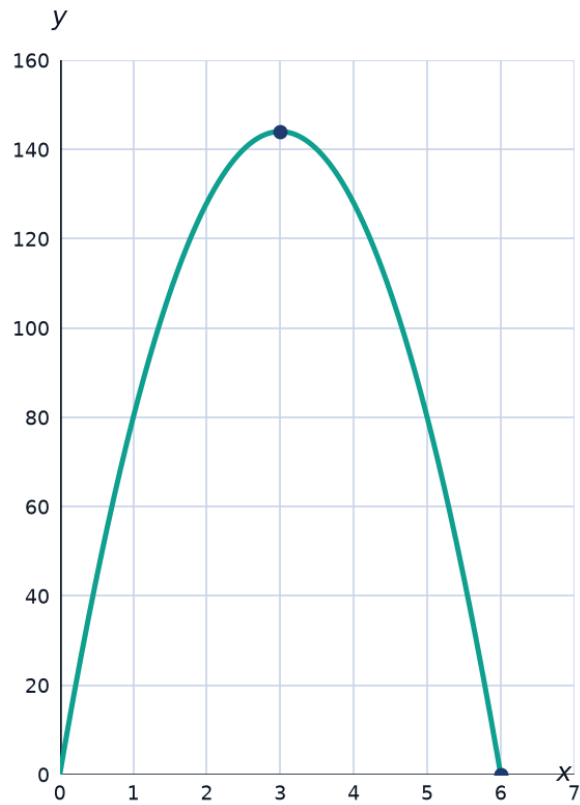
Converting between standard, factored, and vertex forms

- 17 A parabola is given in factored form as $y = 2(x - 1)(x + 3)$. Convert this to standard form.
- 18 A parabola has zeros at $x = 2$ and $x = 6$, and passes through $(4, -8)$. Find its equation in factored form.



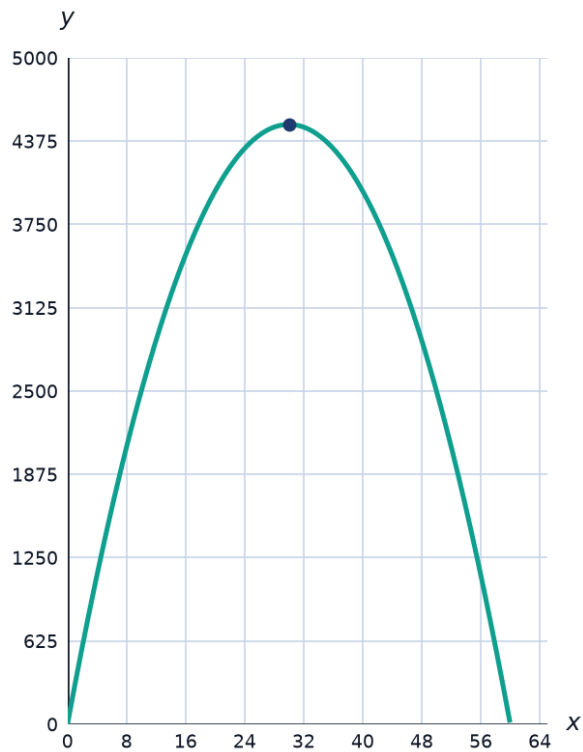
Quadratic word problems: projectile motion

- 19 A rocket is launched with height modeled by $h(t) = -16t^2 + 96t$, where t is time in seconds. Find when the rocket hits the ground, and find its maximum height.



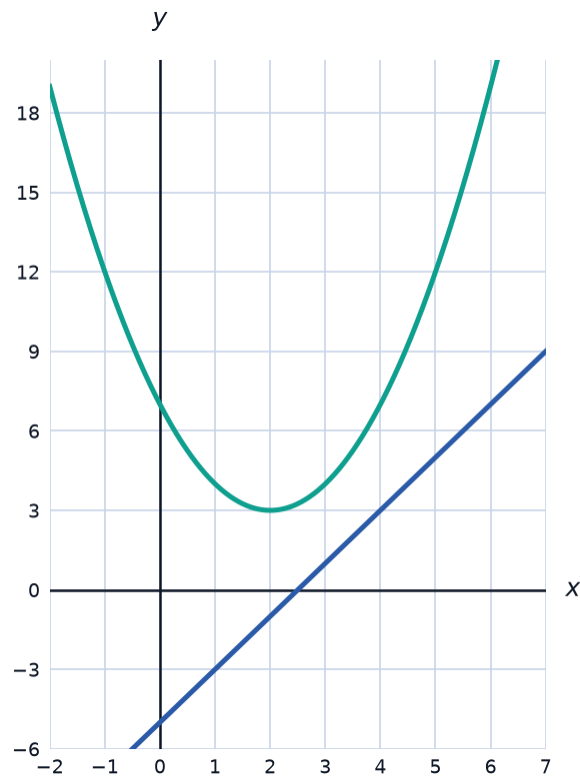
Quadratic optimization in business contexts

- 20 A company's revenue is modeled by $R(x) = -5x^2 + 300x$, where x is the price per item in dollars. Find the price that maximizes revenue, and find the maximum revenue.



Determining the number of solutions graphically

- 21 A parabola $y = x^2 - 4x + 7$ and a line $y = 2x - 5$ are graphed together. Use the discriminant of the resulting equation to determine how many times the graphs intersect.



Multiple choice — easy 15

- 22 Solve $x^2 + 5x = 0$ by factoring.

- a. $(-4, 0)$
- b. $(5, 0)$
- c. $(-5, 0)$
- d. $(-5, -1)$

- 23 Solve $x^2 - 1 = 0$ by factoring.
- a. $(-1, 1)$
 - b. $(1, -1)$
 - c. $(-1, 0)$
 - d. $(0, 1)$
- 24 Solve $x^2 - 7x - 44 = 0$ by factoring.
- a. $(-4, 11)$
 - b. $(-4, 10)$
 - c. $(4, -11)$
 - d. $(-3, 11)$
- 25 Solve $x^2 - 10x + 9 = 0$ by factoring.
- a. $(2, 9)$
 - b. $(1, 8)$
 - c. $(-1, -9)$
 - d. $(1, 9)$
- 26 Solve $x^2 + 16x + 55 = 0$ by factoring.
- a. $(-11, -6)$
 - b. $(-11, -5)$
 - c. $(-10, -5)$
 - d. $(11, 5)$

27 Solve $x^2 + 14x + 33 = 0$ by factoring.

a. (11, 3)

b. (-11, -3)

c. (-10, -3)

d. (-11, -4)

28 Solve $x^2 - 8x + 15 = 0$ by factoring.

a. (3, 4)

b. (-3, -5)

c. (4, 5)

d. (3, 5)

29 Solve $x^2 - 9x - 36 = 0$ by factoring.

a. (3, -12)

b. (-2, 12)

c. (-3, 12)

d. (-3, 11)

30 Solve $x^2 + 8x + 12 = 0$ by factoring.

a. (-5, -2)

b. (-6, -2)

c. (6, 2)

d. (-6, -3)

31 Solve $x^2 + 8x - 48 = 0$ by factoring.

- a. $(-12, 4)$
- b. $(-11, 4)$
- c. $(-12, 3)$
- d. $(12, -4)$

32 Solve $x^2 - 3x = 0$ by factoring.

- a. $(1, 3)$
- b. $(0, 3)$
- c. $(0, -3)$
- d. $(0, 2)$

33 Solve $x^2 - 49 = 0$ by factoring.

- a. $(-6, 7)$
- b. $(-7, 7)$
- c. $(-7, 6)$
- d. $(7, -7)$

34 Solve $x^2 - 8x + 12 = 0$ by factoring.

- a. $(2, 5)$
- b. $(3, 6)$
- c. $(-2, -6)$
- d. $(2, 6)$

35 Solve $x^2 + 12x + 11 = 0$ by factoring.

a. $(-11, -1)$

b. $(-11, -2)$

c. $(11, 1)$

d. $(-10, -1)$

36 Solve $x^2 - 16 = 0$ by factoring.

a. $(-4, 4)$

b. $(-4, 3)$

c. $(4, -4)$

d. $(-3, 4)$

Multiple choice — medium 15

37 Find the vertex of the parabola $f(x) = 2x^2 - 4x - 6$.

a. $(1, -8)$

b. $(-1, -8)$

c. $(2, -8)$

d. $(1, 8)$

38 Find the vertex of the parabola $f(x) = -3x^2 - 60x - 6$.

a. $(-9, 294)$

b. $(-10, -294)$

c. $(10, 294)$

d. $(-10, 294)$

- 39 Find the vertex of the parabola $f(x) = x^2 - 8x - 6$.
- a. $(4, -22)$
 - b. $(5, -22)$
 - c. $(4, 22)$
 - d. $(-4, -22)$
- 40 Find the vertex of the parabola $f(x) = -x^2 - 12x - 5$.
- a. $(-5, 31)$
 - b. $(-6, -31)$
 - c. $(-6, 31)$
 - d. $(6, 31)$
- 41 Find the vertex of the parabola $f(x) = -3x^2 - 18x - 9$.
- a. $(-2, 18)$
 - b. $(-3, -18)$
 - c. $(-3, 18)$
 - d. $(3, 18)$
- 42 Find the vertex of the parabola $f(x) = x^2 + 16x + 6$.
- a. $(-8, -58)$
 - b. $(8, -58)$
 - c. $(-7, -58)$
 - d. $(-8, 58)$

- 43 Find the vertex of the parabola $f(x) = -x^2 + 2x - 2$.
- a. $(-1, -1)$
 - b. $(1, -1)$
 - c. $(2, -1)$
 - d. $(1, 1)$
- 44 Find the vertex of the parabola $f(x) = 3x^2 - 36x - 7$.
- a. $(6, 115)$
 - b. $(7, -115)$
 - c. $(6, -115)$
 - d. $(-6, -115)$
- 45 Find the vertex of the parabola $f(x) = -3x^2 - 18x - 15$.
- a. $(-3, 12)$
 - b. $(3, 12)$
 - c. $(-3, -12)$
 - d. $(-2, 12)$
- 46 Find the vertex of the parabola $f(x) = -3x^2 - 60x - 15$.
- a. $(-10, 285)$
 - b. $(10, 285)$
 - c. $(-10, -285)$
 - d. $(-9, 285)$

47 Find the vertex of the parabola $f(x) = -x^2 + 6x - 11$.

a. $(-3, -2)$

b. $(4, -2)$

c. $(3, -2)$

d. $(3, 2)$

48 Find the vertex of the parabola $f(x) = x^2 - 8x - 10$.

a. $(-4, -26)$

b. $(5, -26)$

c. $(4, -26)$

d. $(4, 26)$

49 Find the vertex of the parabola $f(x) = x^2 - 14x - 2$.

a. $(-7, -51)$

b. $(7, 51)$

c. $(7, -51)$

d. $(8, -51)$

50 Find the vertex of the parabola $f(x) = -x^2 + 6x + 11$.

a. $(3, -20)$

b. $(-3, 20)$

c. $(3, 20)$

d. $(4, 20)$

51 Find the vertex of the parabola $f(x) = -x^2 - 12x - 10$.

- a. $(-5, 26)$
- b. $(-6, -26)$
- c. $(-6, 26)$
- d. $(6, 26)$

Multiple choice — hard 15

52 A rocket is launched with height modeled by $h(t) = -16t^2 + 288t$, where t is time in seconds. Find the time at which it reaches its maximum height, and that maximum height, as an ordered pair (time, height).

- a. $(10, 1296)$
- b. $(9, 1312)$
- c. $(9, 1296)$
- d. $(18, 1296)$

53 A rocket is launched with height modeled by $h(t) = -16t^2 + 64t$, where t is time in seconds. Find the time at which it reaches its maximum height, and that maximum height, as an ordered pair (time, height).

- a. $(4, 64)$
- b. $(2, 80)$
- c. $(2, 64)$
- d. $(3, 64)$

- 54 A rocket is launched with height modeled by $h(t) = -16t^2 + 128t$, where t is time in seconds. Find the time at which it reaches its maximum height, and that maximum height, as an ordered pair (time, height).
- a. (4, 256)
 - b. (4, 272)
 - c. (5, 256)
 - d. (8, 256)
- 55 A rocket is launched with height modeled by $h(t) = -16t^2 + 256t$, where t is time in seconds. Find the time at which it reaches its maximum height, and that maximum height, as an ordered pair (time, height).
- a. (9, 1024)
 - b. (8, 1040)
 - c. (16, 1024)
 - d. (8, 1024)
- 56 A rocket is launched with height modeled by $h(t) = -16t^2 + 224t$, where t is time in seconds. Find the time at which it reaches its maximum height, and that maximum height, as an ordered pair (time, height).
- a. (8, 784)
 - b. (7, 784)
 - c. (14, 784)
 - d. (7, 800)
- 57 Solve $x^2 - 14x + 42 = 0$ using the quadratic formula, giving exact answers.
- a. $7 \pm \sqrt{7}$
 - b. $7 \pm 2\sqrt{7}$
 - c. $-7 \pm \sqrt{7}$
 - d. $8 \pm \sqrt{7}$

58 Solve $x^2 + 6x + 3 = 0$ using the quadratic formula, giving exact answers.

a. $-3 \pm 2\sqrt{6}$

b. $-3 \pm \sqrt{6}$

c. $3 \pm \sqrt{6}$

d. $-2 \pm \sqrt{6}$

59 Solve $x^2 + 10x + 15 = 0$ using the quadratic formula, giving exact answers.

a. $-4 \pm \sqrt{10}$

b. $5 \pm \sqrt{10}$

c. $-5 \pm 2\sqrt{10}$

d. $-5 \pm \sqrt{10}$

60 Solve $x^2 - 8x + 3 = 0$ using the quadratic formula, giving exact answers.

a. $5 \pm \sqrt{13}$

b. $4 \pm \sqrt{13}$

c. $-4 \pm \sqrt{13}$

d. $4 \pm 2\sqrt{13}$

61 Solve $x^2 - 2x - 17 = 0$ using the quadratic formula, giving exact answers.

a. $-1 \pm 3\sqrt{2}$

b. $1 \pm 4\sqrt{2}$

c. $1 \pm 3\sqrt{2}$

d. $2 \pm 3\sqrt{2}$

- 62 A parabola has zeros at $x = 2$ and $x = -5$, and passes through $(-3, -30)$. Find the leading coefficient a in its factored form $y = a(x - (2))(x - (-5))$.
- a. 3
 - b. 4
 - c. 2
 - d. -3
- 63 A parabola has zeros at $x = 3$ and $x = -2$, and passes through $(-6, -36)$. Find the leading coefficient a in its factored form $y = a(x - (3))(x - (-2))$.
- a. -2
 - b. -1
 - c. 1
 - d. 0
- 64 A parabola has zeros at $x = -6$ and $x = 3$, and passes through $(6, 108)$. Find the leading coefficient a in its factored form $y = a(x - (-6))(x - (3))$.
- a. 3
 - b. 2
 - c. -3
 - d. 4
- 65 A parabola has zeros at $x = -2$ and $x = 6$, and passes through $(2, 32)$. Find the leading coefficient a in its factored form $y = a(x - (-2))(x - (6))$.
- a. -2
 - b. 2
 - c. -1
 - d. -3

- 66 A parabola has zeros at $x = -1$ and $x = -2$, and passes through $(-8, -84)$. Find the leading coefficient a in its factored form $y = a(x - (-1))(x - (-2))$.
- a. 2
 - b. -3
 - c. -2
 - d. -1